Notes on Radio Evolution of Galaxies

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PRL '26

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Chapter 1

Useful Terminology and Concepts

TODO

- add section for atomic and molecular spectral lines, H alpha, cool gas, all of that stuff.

1.1 Basic Terminology

need to add magnitude concise defns, colour index followed.

i. add parsec defn

ii. **Luminosity** (L): The total amount of electromagnetic **energy emitted per unit time** by an astronomical object.

iii. **Radiant Flux Density** (S): The total amount of electromagnetic **energy per unit area per unit time** by an astronomical object. In other words, it is the luminosity per unit area. According to this definition, at a distance D from a star (say):

$$S = \frac{L}{4\pi D^2}$$

iv. **Effective Temperature** (T_{eff}): The effective temperature of a body such as a star or planet is the temperature of a black body of the same radius that would emit the same total energy as electromagnetic radiation.

v. **Colour Temperature** (T_c): The temperature of a black body such that it has the same colour index as that of an astral body.

vi. **Specific Luminosity** (L_ν): The energy emitted per unit time for a unit frequency range.

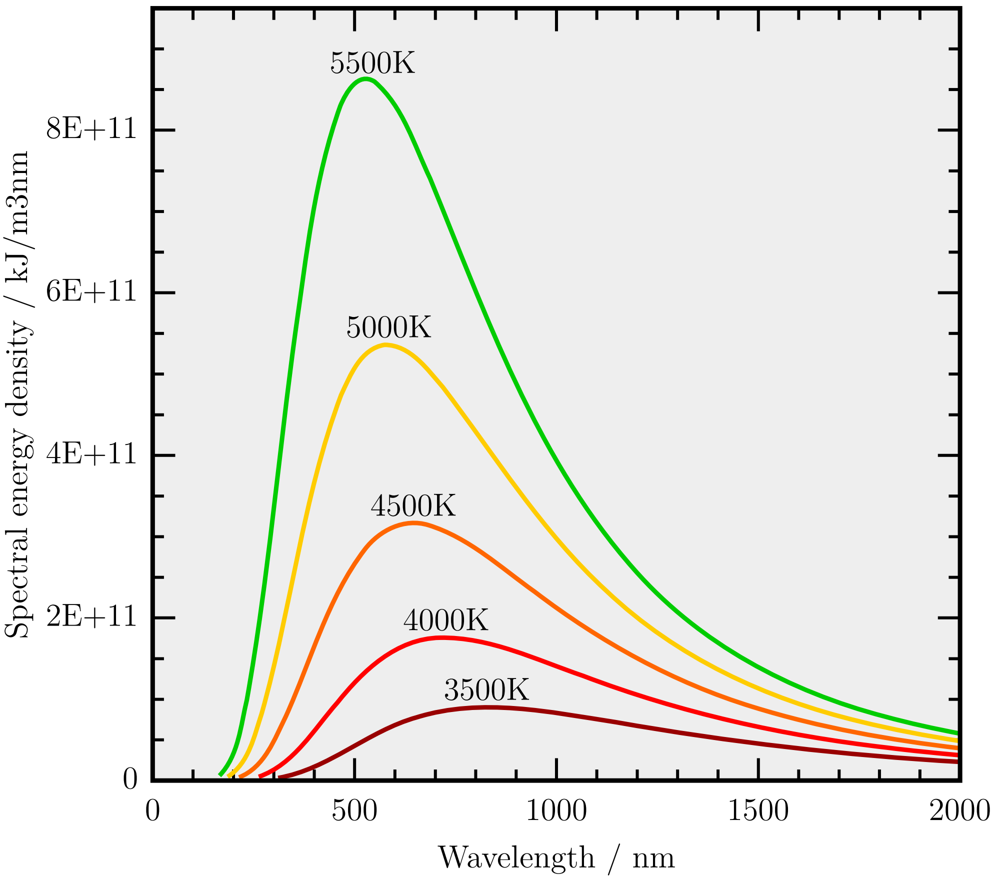
vii. add sun units, $\odot$

1.2 Blackbody Radiation

A very brief explanation of blackbody radiation includes the Stefan-Boltzmann Law which states that the flux S for a perfect blackbody is given by:

$$S = \sigma T^4 \tag{1.1}$$

where $\sigma = 5.67 \times 10^{-5}$ erg cm^{-2} K^{-4} s^{-1} is the Stefan-Boltzmann constant.



Additionally, for a blackbody, it helps to know the Wien's Displacement Law which states that the emission curve for blackbodies peak at different wavelengths which is inversely proportional to its temperature:

$$\lambda_{\text{peak}} \propto \frac{1}{T} \tag{1.2}$$

1.3 The Magnitude Scale

Our eyes judge brightness according in a logarithmic fashion, similar to hearing. This means that what we perceive to be twice as bright is actually emitting a lot more than just twice the radiant energy. Due to this, optical astronomy uses 'magnitude' as a scale of brightness.

1.3.1 Apparent Magnitude

The purpose of apparent magnitude is to closely match visually reported magnitudes and quantify these historical observations. It is a measure for the radiant flux S from a body. For two sources with fluxes S_1 and S_2, their *apparent magnitudes* m_1 and m_2 are defined as:

$$m_1 - m_2 = -2.5 \log\left(\frac{S_1}{S_2}\right)$$

This means that a brighter object will have a smaller apparent magnitude. The factor of 2.5 is chosen to most accurately match the historically reported magnitudes. A more *absolute* definition can be found below in the next section, which depends on the filter in consideration.

1.3.2 Filters and Colour

Optical observations are performed with a combination of filters and detectors. Since the spectral energy distribution is not necessarily uniform over all wavelengths, apparent magnitudes are defined for each of these filters, i.e. apparent magnitude is a function of the wavelength of light being considered.

Vega Magnitude

$$m = m(\lambda) \equiv m_X \equiv X$$

where X is the filter for some specific wavelength λ.

The magnitudes of the different filters are related, by definition, as $U = B = V = R = I = \dots$ for main sequence stars of spectral type A0 (Vega is an archetype of such stars), i.e., every colour index for such a star is defined to be zero.

For a more precise definition, let $T_X(\nu)$ be the transmission curve of the filter-detector system. $T_X(\nu)$ specifies which fraction of the incoming photons with frequency are registered by the detector. The apparent magnitude of a source with spectral flux S_ν is then:

$$m_X = -2.5 \log\left(\frac{\int \mathrm{d}\nu \; T_X(\nu) S_\nu}{\int \mathrm{d}\nu \; T_X(\nu)}\right) + \text{const.}$$

The argument of the logarithm is the average flux received in total for the entire EM spectrum. The constant is determined from reference stars. This is also referred to as the Vega apparent magnitude.

AB Magnitude

Another commonly used definition, does not consider the spectral energy distribution of reference. The reference is instead defined as constant flux at all frequencies, $S_\nu^{\rm ref} = S_\nu^{\rm AB} = 2.89 \times 10^{-21}$ erg s^{-1} cm^{-2} Hz^{-1}. This value makes A0 stars like Vega have the same magnitude in the original Johnson-V-band as in the AB system.

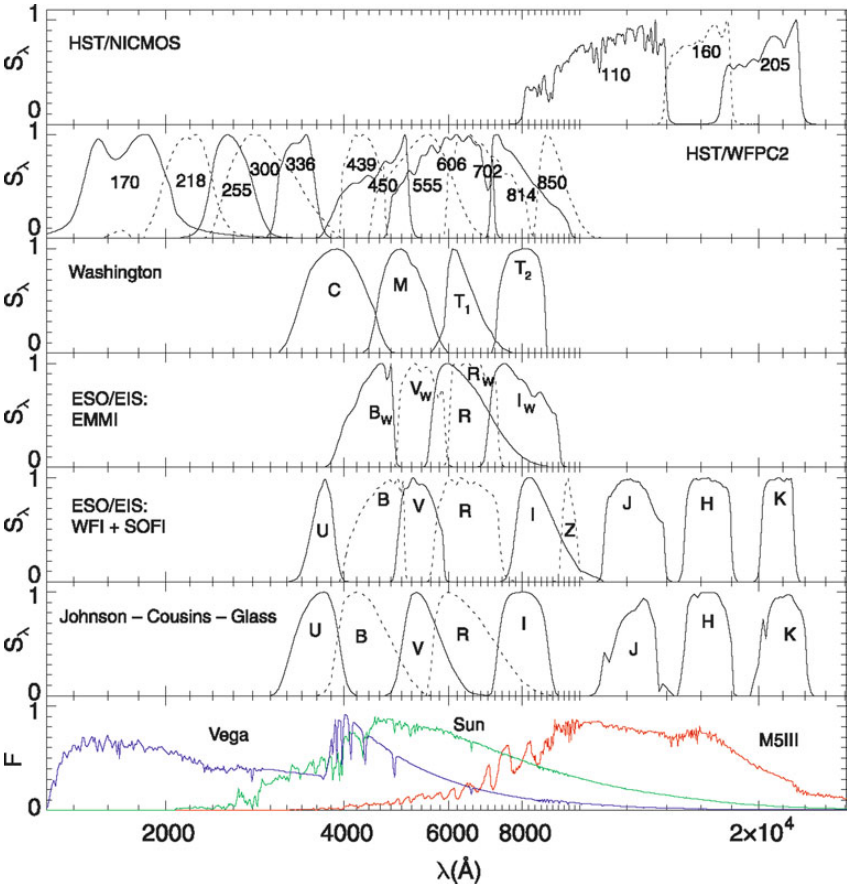


Figure 1.1: This shows the transmission curves of different spectra. At the bottom, we have the spectra of 3 stars with different effective temperatures.

1.3.3 Absolute Magnitude

Since apparent magnitude does not reveal any information about the luminosity of the body since it inherently depends on the distance of the body from Earth. If we are at a distance D from a body and L_ν (energy per unit time for unit frequency range) is its specific luminosity, the flux for isotropic emission is given by:

$$S_\nu = \frac{L_\nu}{4\pi D^2} \tag{1.3}$$

To reveal the physical properties of the source itself irrespective of how it appears to us, we define absolute magnitude, M_X, where X refers to the filter under consideration. It is defined to be the apparent magnitude of a body at a distance of 10 pc from us. Its relation to the apparent magnitude is as follows:

$$m_X - M_X = 5\log\left(\frac{D}{1pc}\right) - 5 \equiv \mu$$

where μ is called the *distance modulus*. Hence, the latter is a logarithmic measure of the distance of a source: $\mu = 0$ for $D = 10$ pc, $\mu = 10$ for $D = 1$ kpc, and $\mu = 25$ for $D = 1$ Mpc. The difference between apparent and absolute magnitude is independent of the filter choice, and it equals the distance modulus if no extinction is present. In general, this difference is modified by the wavelength (and thus filter) dependent extinction coefficient.

1.3.4 Bolometric Parameters

The total flux S and luminosity L over all frequencies are naturally be defined as:

$$S = \int_0^\infty S_\nu \, \mathrm{d}\nu; \qquad L = \int_0^\infty L_\nu \, \mathrm{d}\nu$$

These define the *bolometric magnitudes*:

$$\begin{aligned} m_{\mathrm{bol}} &= -2.5\log(S) + \mathrm{const.} \\ M_{\mathrm{bol}} &= -2.5\log(L) + \mathrm{const.} \end{aligned}$$

The constants can be determined from reference stars. In particular, using the sun as reference with $m_{\odot\mathrm{bol}} = -26.83$ and the distance of 1 a.u. gives $\mu = -31.47$, thus giving us:

$$M_{\odot\mathrm{bol}} = m_{\odot\mathrm{bol}} - \mu = 4.74$$

From this, we obtain:

$$M_{\mathrm{bol}} = 4.74 - 2.5\log\left(\frac{L}{L_\odot}\right)$$

where the luminosity of the sun is $L_\odot = 3.85 \times 10^{33}$ erg s^{-1}. To obtain a value for the luminosity for galaxies or AGNs[1], we use the relative definition of absolute magnitude to write:

$$L_X = 10^{-0.4(M_X - M_{\odot X})} L_{\odot X} \tag{1.4}$$

where X represents the X-band filter.

[1] Active Galactic Nuclei (AGN) are galaxies whose largest fraction of luminosity is produced in a very small central region: the active galactic nucleus.

1.4 Stars

Stars can be reasonably thought of as gas spheres, in the cores of which light atomic nuclei fuse into heavier ones (mostly hydrogen to helium) by thermonuclear processes, thereby producing energy. **External appearance of stars** is primarily determined by their radii R and their temperature T. **Properties and evolution of stars** is primarily governed by their masses M.

1.4.1 Parameters of Stars

For a first approximation, the spectral energy distribution of a star's emission can be described by blackbody radiation, where the flux S is given by the Stefan-Boltzmann equation (1.1). Thus, the luminosity for the star at a distance R is given by (1.3):

$$L = 4\pi R^2 \ \sigma T^4$$

Stars deviate from this profile and we define effective temperature (iv.) as:

$$\sigma T_{\text{eff}}^4 \equiv \frac{L}{4\pi R^2} \tag{1.5}$$

The luminosities can range from $10^{-4} L_\odot$ to $10^5 L_\odot$.

The temperature of a star can be estimated from its colour. Again, to a first approximation, Wien's displacement law (1.2) can explain the colour of stars depending upon temperature. mention how redshift gets factored in here, found relevant thread here.

Using the colour index $X - Y \equiv m_X - m_Y$ in two filters X and Y lets us find the colour temperature T_c (v.) of the star. For a perfect blackbody, $T_c = T_{\text{eff}}$ but in general they differ.

1.4.2 Classification of Stars

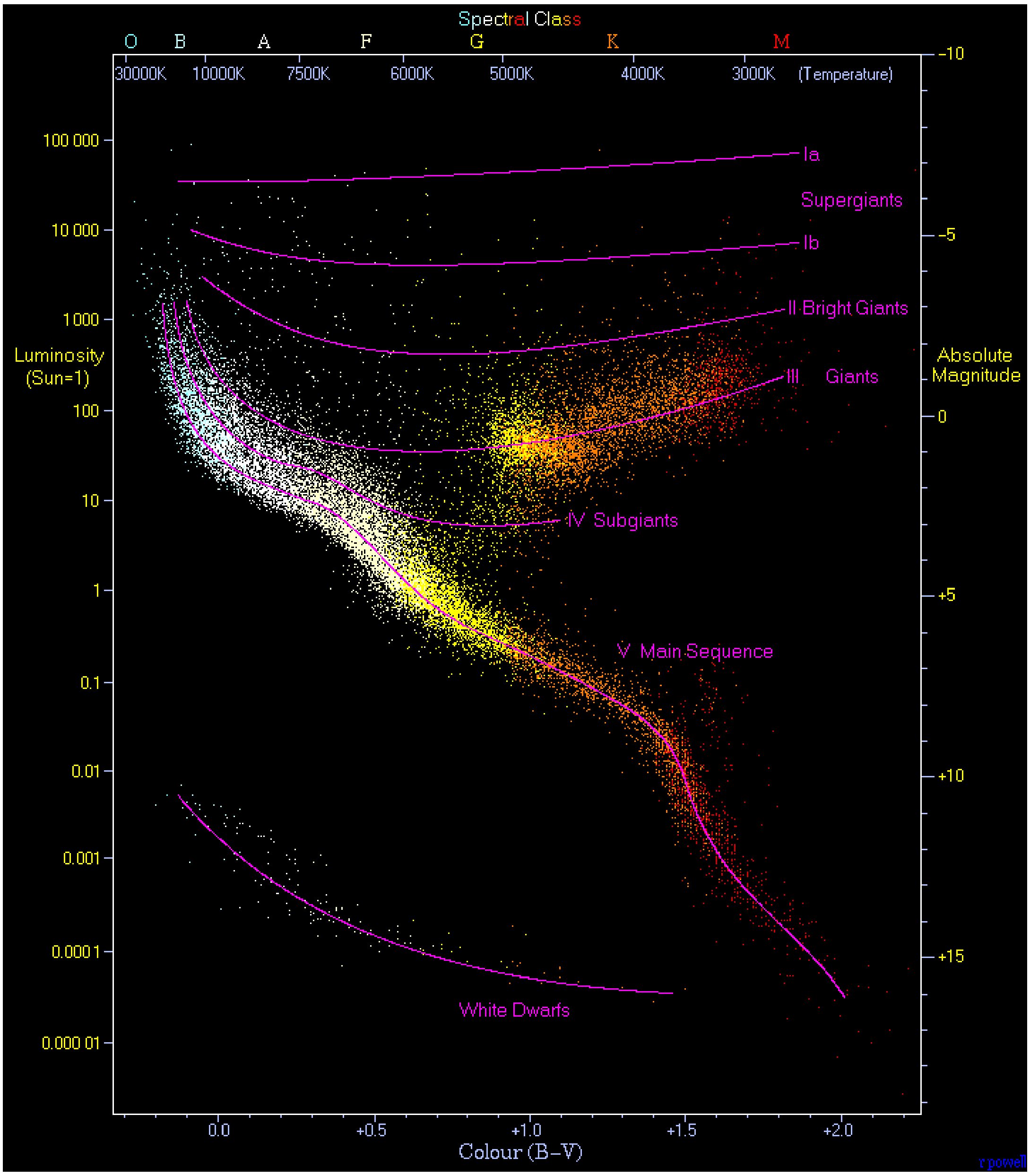
The Harvard sequence of stellar spectra classifies stars according to the **widths and ratios of their atomic spectral lines**. This is essentially a classification based on **colour temperature** T_c since the transitions leading to those spectral lines are directly dependent on the temperature of the region.

The sequence of **spectral classes** is denoted in the order O, B, A, F, G, K, M in decreasing order of colour temperature. Each spectral class is further subclassified using digits 0 to 9, again in descending order of colour temperature. For instance, our Sun is a G2 star.

Hertzsprung-Russell Diagram (HRD): Absolute magnitude vs. spectral classes.

Colour-Magnitude Diagram (CMD): Absolute magnitude vs. colour index (like $B - V$ or $V - I$)

Both of these diagrams yield a very charcteristic central result of astronomy which says that most stars are arranged along a 1-dimensional sequence - the main sequence. Luminosity L vs effective temperature $T_{\rm eff}$ also yields a different but very similar graph.



Based on luminosity, the stars can be classified into **luminosity classes**:

Class I Supergiants; Class II Bright giants

Class III Giants; Class IV Subgiants

Class V Dwarfs; Class VI Subdwarfs

Our sun is a luminosity class V dwarf.

Since stars exist which have, for the same spectral type and hence the same color temperature (and roughly the same effective temperature), very different luminosities, we can deduce immediately that these stars have different radii from the equation of effective temperature (1.5). Thus the red giants which have the same effective temperatures as the main sequence stars must have much larger radii.

Spectroscopically, the width of spectral absorption lines depends on gravitational acceleration on star's surface, $g = \frac{GM}{R^2}$, according to models of stellar atmosphere. It is based on these line widths (and thus luminosity) that we obtain the different luminosity classes.

If we have the distance of a star (thus its luminosity) and if its surface gravity can be derived from the line width, we can obtain the **stellar mass**. For the **main sequence stars**, this turns out to be approximately:

$$\frac{L}{L_\odot} \approx \left(\frac{M}{M_\odot}\right)^{3.5} \tag{1.6}$$

1.4.3 Spectra of Some Stars

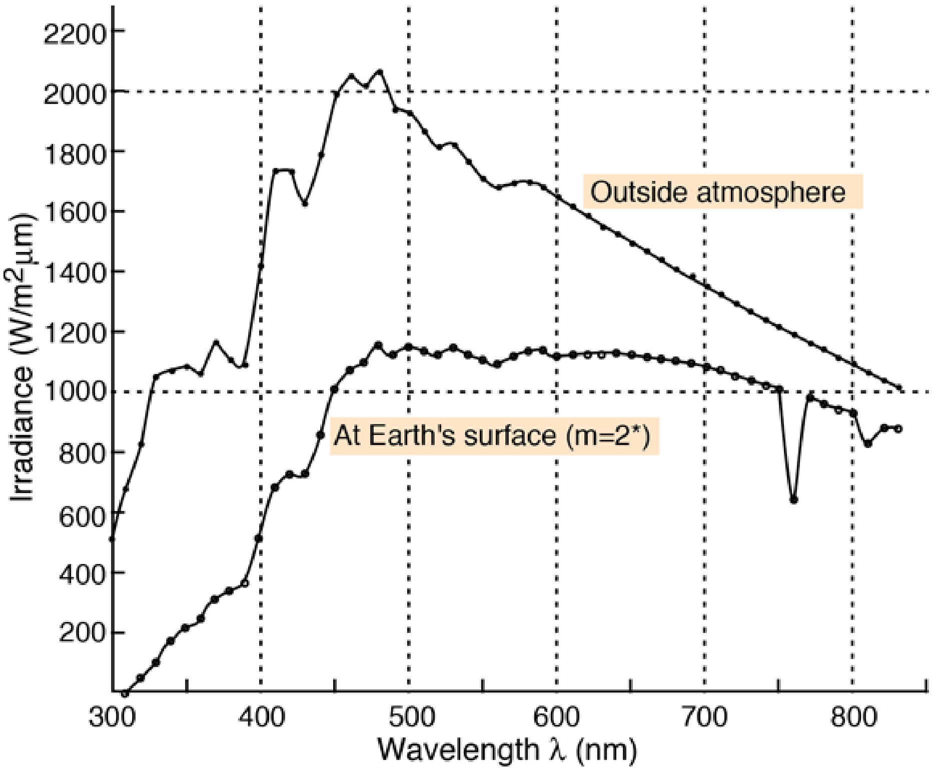


Figure 1.2: Solar Spectrum, G2 Type Star

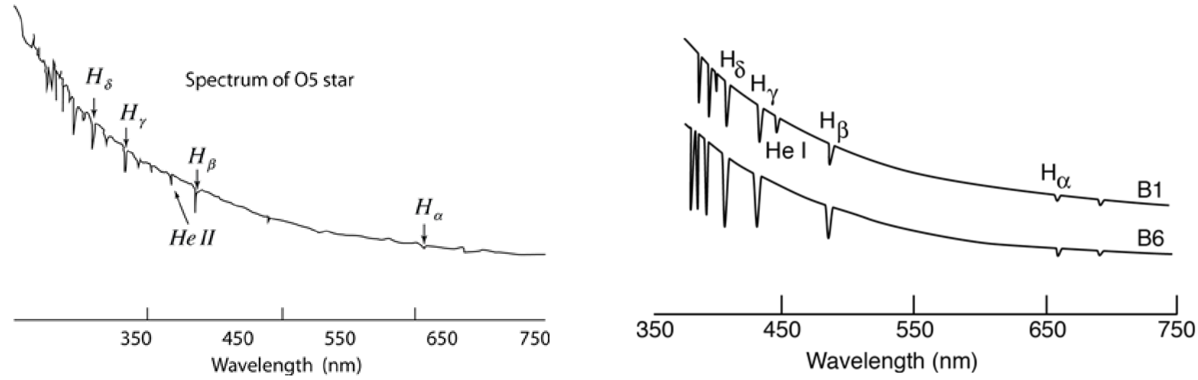


Figure 1.3: Spectrum of an O5 Type Star (left) and B Type Stars (right).

1.4.4 Structure and Evolution of Stars

Stars are almost spherically symmetric, so we consider its parameters as radial functions: Pressure, Density, Temperature and Chemical Composition. Stars are almost always in hydrostatic equilibrium, with the pressure fighting against its own gravitational pull.

Fusion at the Center and Transport of Energy

In the center of a star, the density and temperature are high enough to cause thermonuclear ignition, causing the formation of Helium from Hydrogen (4 protons combine into ^{4}He nucleus). This fusion results in energy release that is transported outwards in the form of radiation transport or through convection of stellar plasma. It is then released in the form of electromagnetic waves from the star. The chemical composition is homogenous in the convection zones as they transport and mix the stellar material.

A part of the energy released is in form of neutrinos that can escape unobstructed from stars because of their low cross-section[2].

Temperature Dependence of Energy Production Rate

For temperatures below around 15×10^6 K, the energy production rate is approximately proportional to T^4, where the reaction follows the 'pp-chain'. Above that, the reaction follows the 'CNO cycle' where the rate is roughly proportional to T^{20}.

Lifetime of a Star

Beginning with homogenous chemical composition, if the stars have masses above $\sim 0.08 M_\odot$, only then do they have enough temperature and pressure in their core to fuse

[2] Cross-section is a measure related to the probability that a specific process will take place in a collision of two particles. In this case, it is referring to the probability that a neutrino will interact with matter.

hydrogen into helium. Otherwise, they are called brown dwarfs which aren't exactly stars in the proper sense.

At the onset of nuclear fusion, the star is located on the Zero-Age Main Sequence (ZAMS) in the HRD. With the conversion of hydrogen into helium, the chemical composition changes and the conditions in a star will change considerably once roughly 10% of its hydrogen is used up. At that point, the stars will move away from the main sequence. This allows us to estimate the life of a star on the main sequence. The total energy produced in this phase is:

$$E_{\mathrm{MS}} = 0.1 \times M\ c^2 \times 0.007$$

where $M\ c^2$ is the rest mass of the star and 0.007 is the efficiency of the conversion. We know that luminosity is energy per unit time, so this energy is equivalent to the product of the luminosity in this period with the time period t_{MS}:

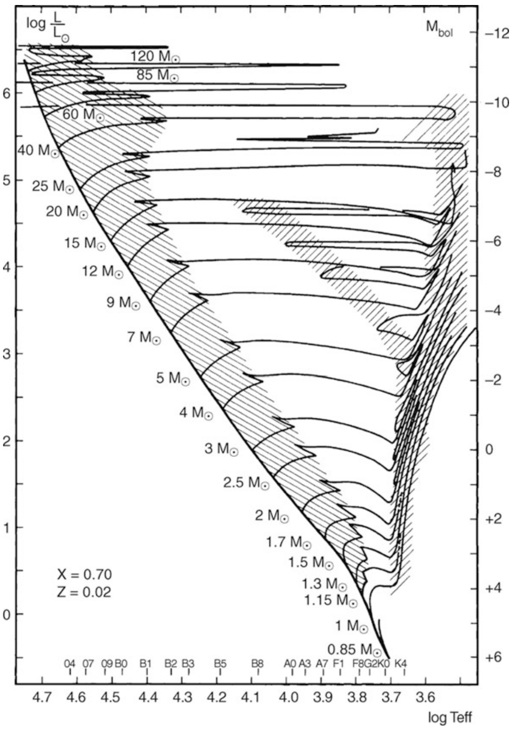


Figure 1.4: The solid line represents the ZAMS in this temperature-luminosity diagram.

$$t_{\mathrm{MS}} = \frac{E_{\mathrm{MS}}}{L} \approx 8 \times 10^9\ \frac{M/M_{\odot}}{L/L_{\odot}}\ \mathrm{yr} \approx 8 \times 10^9\ \left(\frac{M}{M_{\odot}}\right)^{-2.5}\ \mathrm{yr}$$

where we used the fact that the luminosity is a steep function of the stellar mass (1.6).

Thus, heavier the star, lesser the time it spends on the main sequence. A G star like our sun stays in the main sequence for about 8-10 billion years, whereas an O or B star remains only for a few million years. In the course of their evolution on the main sequence, the stars move away slightly from the ZAMS in the HRD towards somewhat higher luminosities and lower effective temperatures (so towards down right in the diagram). Massive stars can lose part of their initial mass by stellar winds.

Evolution Post Main Sequence

Evolution post the initial phase depends on the stellar mass. Suppose the initial mass of the star is M.

- For $M \leq 0.7M_\odot$

  The initial hydrogen burning phase[3] of such stars is longer than the (estimated) age of the universe[4]. These stars still lie on the main sequence.

- For $M \leq 2.5M_\odot$

  Central hydrogen burning is followed by a relatively brief phase of the hydrogen fusing into helium in a shell outside the center of the star. After this phase, the star moves right in the HRD (lower effective temperature but same luminosity) and thereby strongly expands (1.5). This causes the temperature and density in the center rise enough so that helium starts fusing into carbon.

  This helium burning is very explosive for these stars, resulting in 'helium flashes', during which large amounts of stellar mass get ejected. After these flashes, a new stable equilibrium configuration is established with a helium burning shell formed besides the hydrogen burning shell. Expanding its radius, the star will evolve into a red giant or supergiant and move along the asymptotic giant branch (AGB) in the HRD.

- For $M \leq 8M_\odot$

  After the initial expansion due to the formation of the hydrogen burning shell, these stars have their cores hot and dense enough for 'quiet' helium formation. Once the core runs out of helium, a second shell source forms fusing helium. In this stage, the star becomes a red giant or supergiant. A large portion of stellar mass is ejected into the interstellar medium in the form of stellar winds.

  The configuration of the helium shell is unstable and it releases flashes (helium is burned) in the form of pulses. After some time, this leads to the ejection of the outer envelope[5] which then becomes visible as a *planetary nebula*.

  At this stage, since the star is shedding its outer envelope, its radius decreases drastically, leading to a very drastic increase in temperature (1.5), moving it left in the HRD. The fuel burning in the core is not affected, hence the luminosity stays mostly the same. However, once the fuel runs out, there is a decrease in the luminosity which leads to a small decrease in the temperature until it cools down to become a *white dwarf* with a mass of about $0.6M_\odot$ and a radius roughly the same as that of Earth.

[3] Hydrogen burning just refers to its fusion into helium. Similarly, helium burning refers to its fusion into carbon.

[4] The current estimate for the age of the universe is around 13.787 billion years.

[5] The outer envelope is everything apart from the star's core.

- For $M \geq 8M_\odot$

  For these stars, the temperature and density at its center become so tremendous that carbon can also be fused.Subsequent stellar evolution leads to a core-collapse supernova, leading to the formation of a neutron star, or for the most massive ones ($\sim M \geq 25$), a black hole.

Chapter 2

The World of Galaxies

Ever since the Hubble Deep Field image, the whole of humanity has set their sight to discover what lies far beyond they could ever comprehend. When dealing with extragalactic astronomy, the most notable entities are arguably other galaxies, and this is a chapter all about them.

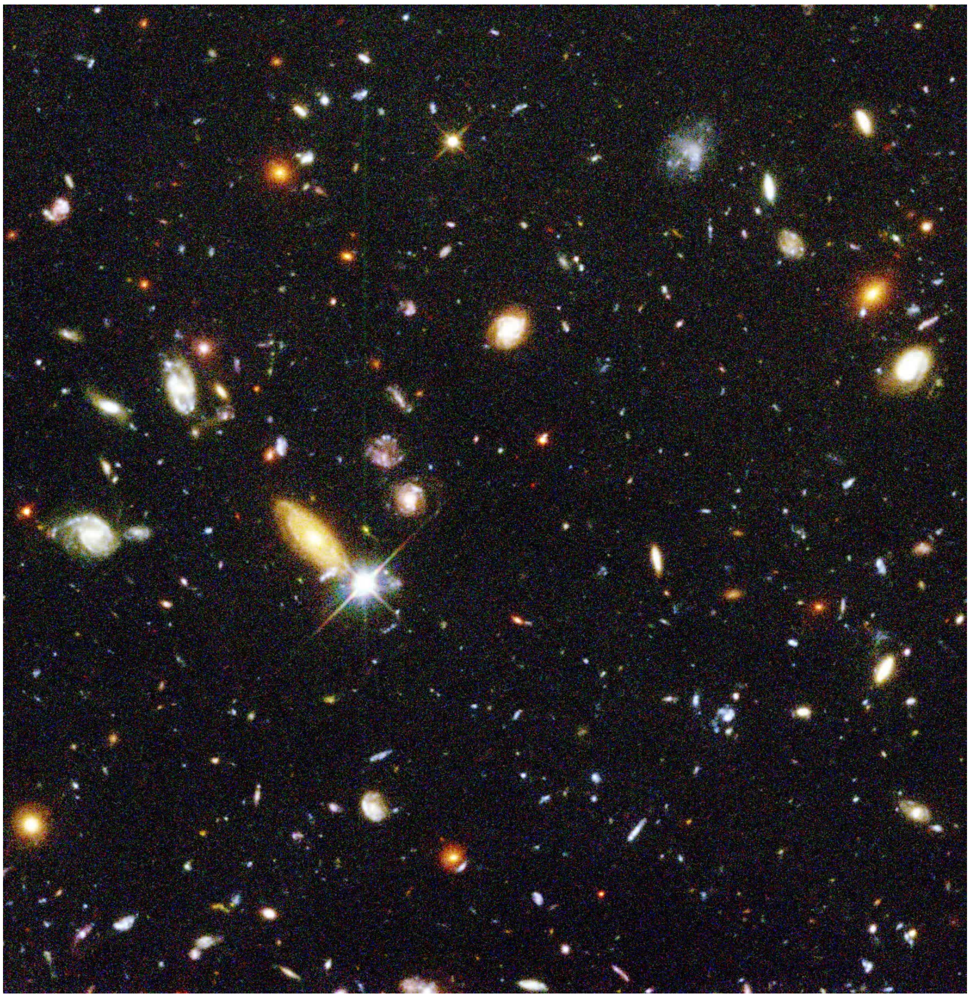


Figure 2.1: The Hubble Deep Field image produced by *Edwin Hubble* changed the world forever. All the dots in the image are galaxies, not stars.

The light from 'normal galaxies' is mainly the sum of all the light emitted by the stars present within it. need to add note about age of stars anol but for that I need to add the section about stars at the beginning.

2.1 Classification of Galaxies

Galaxies dafds

2.1.1 Morphological Classification

On the basis of shape or morphology, galaxies have been historically classified into the following categories:

1. **Elliptical Galaxies**: Their isophotes[1] resemble ellipses. They are further subclassified according to the **ellipticity** of the isophotes: $\epsilon = 1 - b/a$, where a, b are the semi-major and semi-minor axes. Then we define $N = 10\epsilon$ which gives us the representation EN for elliptical galaxies. An E7 galaxy is an elliptical galaxy with $b/a = 0.3$.
2. **Spiral Galaxies**: Their isophotes look like spirals.

[1] Isophotes are contours along which the surface brightness is constant.